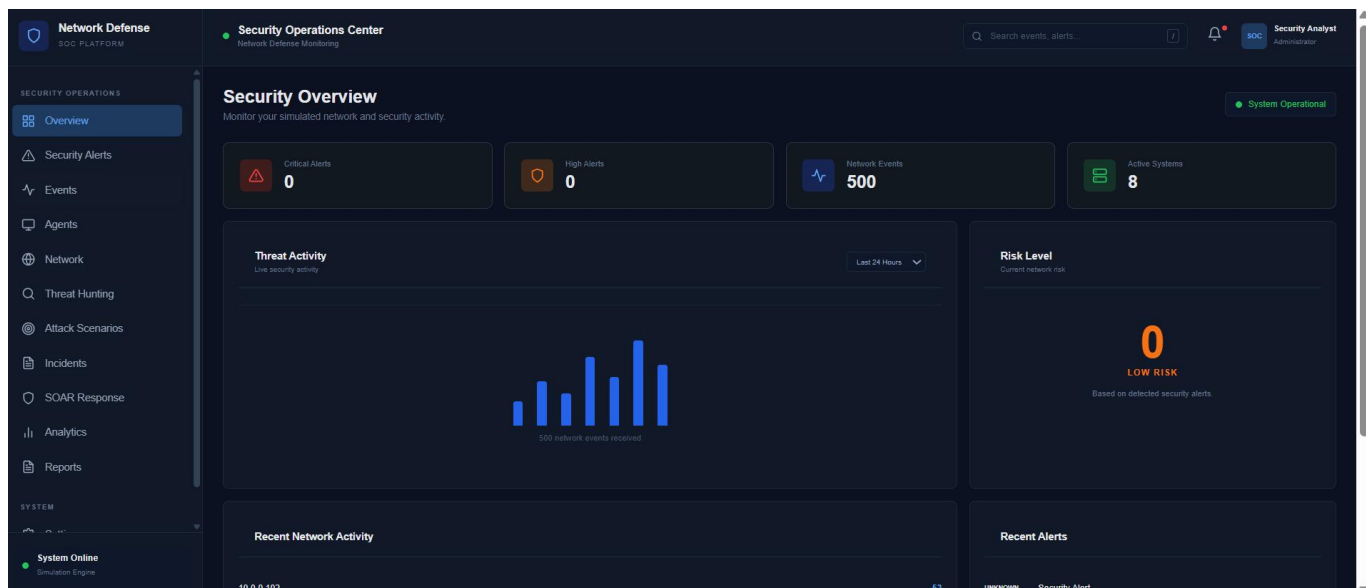


NETWORK DEFENSE SIMULATION

1. Project Overview

The Network Defense Simulation is a learning-oriented SOC platform that demonstrates a security monitoring and incident-response workflow. It generates controlled security activity, represents it as events, applies detection logic, produces alerts, supports incident investigation, provides a simulated SOAR response workflow, and aggregates activity through analytics.

- Frontend: ReactJS
- Backend: Node.js with Express.js
- Communication: REST APIs
- Styling: Plain CSS
- Security concepts: SOC, SIEM, detection rules, incidents, MITRE ATT&CK and SOAR



2. Backend / API Health

The backend provides the REST API used by the frontend. A health check confirms that the Node.js/Express server is online.

SNAPSHOT 1 — Backend / API Health

```
PS C:\Users\ADMIN\network-defense-sim> Invoke-RestMethod http://localhost:5000/

message                status
-----                -
Network Defense Simulation API online
```

Command / action for this screenshot

Invoke-RestMethod http://localhost:5000/

3. Start the Security Simulation

The simulation generates controlled security activity that can be monitored by the SOC interface.

SNAPSHOT 2 — Start the Security Simulation

```
PS C:\Users\ADMIN\network-defense-sim> Invoke-RestMethod http://localhost:5000/api/simulation/status

running totalEvents totalAlerts
-----
True      8          8
```

Command / action for this screenshot

Invoke-RestMethod -Method POST http://localhost:5000/api/simulation/start

Invoke-RestMethod http://localhost:5000/api/simulation/status

4. Events / Security Telemetry

Events represent underlying activity or telemetry generated by the simulation. They are the activity that detection logic can analyze.

SNAPSHOT 3 — Events / Security Telemetry

Event Explorer

Search and investigate raw security events collected from the network.

246 Events

Q

Search IP, protocol, rule, action...

All Severity

TIME	SEVERITY	SOURCE	DESTINATION	PROTOCOL	PORT	ACTION	RULE
9/23/2026, 6:46:13 PM	INFO	Unknown	Unknown	TCP	-	ALLOW	Network Event
9/23/2026, 6:46:14 PM	INFO	Unknown	Unknown	TCP	-	ALLOW	Network Event
9/23/2026, 6:46:15 PM	INFO	Unknown	Unknown	UDP	-	ALLOW	Network Event
9/23/2026, 6:46:16 PM	INFO	Unknown	Unknown	TCP	-	ALLOW	Network Event
9/23/2026, 6:46:17 PM	INFO	Unknown	Unknown	TCP	-	ALLOW	Network Event
9/23/2026, 6:46:18 PM	INFO	Unknown	Unknown	TCP	-	ALLOW	Network Event

Command / action for this screenshot

Invoke-RestMethod <http://localhost:5000/api/events>

5. Security Alerts / Detection

Detection logic evaluates activity and creates security alerts when a suspicious pattern or rule condition is identified.

SNAPSHOT 4 — Security Alerts / Detection

Security Alerts

Monitor, investigate and manage security detections.

Refresh

TOTAL ALERTS

100

CRITICAL

0

HIGH

0

OPEN

100

Search alerts...

All Severities

All Status

SEVERITY	DETECTION	SOURCE	DESTINATION	STATUS	ACTION
UNKNOWN	Security Detection Detection Engine	10.0.0.10	10.0.0.20	OPEN	Investigate
UNKNOWN	Security Detection Detection Engine	10.0.0.10	10.0.0.20	OPEN	Investigate
UNKNOWN	Security Detection Detection Engine	10.0.0.30	10.0.0.10	OPEN	Investigate
UNKNOWN	Security Detection Detection Engine	10.0.0.10	10.0.0.20	OPEN	Investigate
UNKNOWN	Security Detection Detection Engine	10.0.0.101	10.0.0.10	OPEN	Investigate

Command / action for this screenshot

```
$alerts = @(Invoke-RestMethod http://localhost:5000/api/alerts)
$alerts | Select-Object -First 5 id,severity,title,status
```

6. Detection Statistics

Detection statistics provide an aggregated view of events, alerts and detection-rule activity.

SNAPSHOT 5 — Detection Statistics

```
PS C:\Users\ADMIN\network-defense-sim> Invoke-RestMethod http://localhost:5000/api/detection/stats

totalEvents totalAlerts detectionRules
-----
346          100 @{UNKNOWN=477; HIGH_TRAFFIC_VOLUME=23}
```

Command / action for this screenshot

Invoke-RestMethod http://localhost:5000/api/detection/stats

7. Alert Investigation

The analyst reviews an alert to understand severity, source, timestamp, status and detection context before creating an incident.

SNAPSHOT 6 — Alert Investigation

Raw Event JSON

```
{
  "id": "evt-1790169373650-8887",
  "timestamp": "2026-09-23T13:16:13.650Z",
  "profile": "VPN Traffic",
  "sourceIP": "10.0.0.30",
  "sourceDevice": "VPN Server",
  "destinationIP": "10.0.0.10",
  "destinationDevice": "Web Server",
  "protocol": "TCP",
  "destinationPort": 443,
  "service": "HTTPS",
  "bytes": 4125,
  "packets": 14,
  "action": "ALLOW"
}
```

8. Incident Management

An incident represents the tracked investigation created from a security alert. The analyst can review it, add notes, link related alerts and resolve it.

SNAPSHOT 7 — Incident Management

Incident Management

Investigate, track and resolve security incidents.

Total Incidents
1

Open
1

Investigating
0

Resolved
0

Security Incidents

1 incident

ID	TITLE	SEVERITY	STATUS	CREATED	
INC-1790169930241	Security Incident Security event requires investigation.	LOW	OPEN	9/23/2026, 6:55:30 PM	Investigate

Command / action for this screenshot

Create an incident from a CURRENT alert through the UI. API alternative:

```
$incident = Invoke-RestMethod -Method POST -Uri http://localhost:5000/api/incidents -ContentType 'application/json' -Body (@{alertId=$alertId} | ConvertTo-Json)
```

9. Investigation Notes

Analyst notes document observations and investigation progress.

SNAPSHOT 8 — Investigation Notes

INC-1790169930241

Security Incident

×

Severity

LOW

Status

OPEN

Created

9/23/2026, 6:55:30 PM

Description

Security event requires investigation.

Related Alerts

alert-1790169855090

Investigation Notes

SOC Analyst

Initial investigation identified suspicious activity. Reviewing related events and determining appropriate response.

9/23/2026, 6:58:17 PM

Command / action for this screenshot

Add a note to the incident through the UI. API alternative:

```
Invoke-RestMethod -Method POST -Uri "http://localhost:5000/api/incidents/$incidentId/notes" -ContentType 'application/json' -Body (@{note='Initial investigation identified suspicious activity. Reviewing related events and determining appropriate response.'} | ConvertTo-Json)
```

10. MITRE ATT&CK Context

Where implemented mapping provides a match, detection information can be associated with MITRE ATT&CK concepts to describe adversary behavior.

SNAPSHOT 9 — MITRE ATT&CK Context

Security Detection

Security detection generated by the network defense engine. / 100

Alert Details

 Detection Rule Unknown Rule	 Source IP 10.0.0.10
 Destination 10.0.0.20	 Timestamp 2026-09-23T13:41:38.906Z

Detection Analysis

The detection engine identified suspicious activity matching the configured security rule.

This alert should be reviewed by the security analyst to determine whether the activity represents malicious behavior.

MITRE ATT&CK

No MITRE ATT&CK mapping is available for this detection.

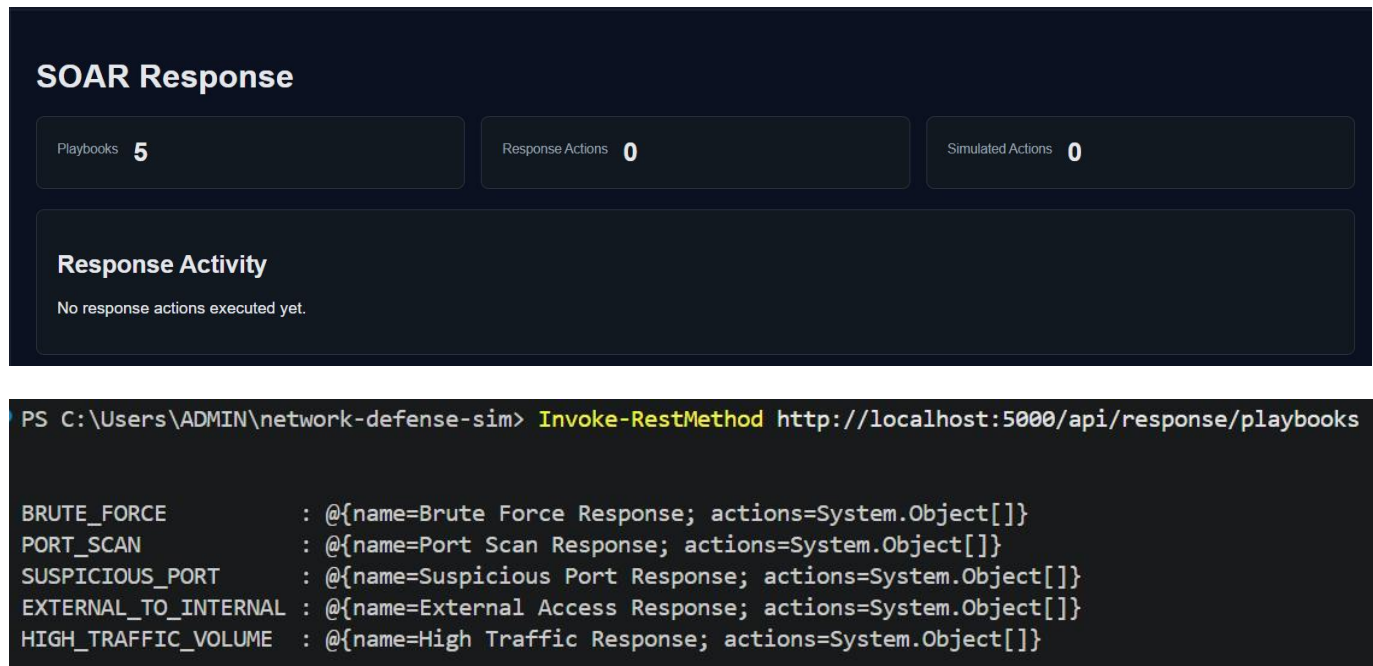
Command / action for this screenshot

Open alert/detection details for a current alert.

11. SOAR Playbooks / Actions

SOAR means Security Orchestration, Automation and Response. In this project the response layer is simulated through playbooks and response actions.

SNAPSHOT 10 — SOAR Playbooks / Actions



The screenshot displays the 'SOAR Response' dashboard and a terminal window. The dashboard shows 5 Playbooks, 0 Response Actions, and 0 Simulated Actions. Below this, the 'Response Activity' section indicates that no response actions have been executed yet. The terminal window shows the command `Invoke-RestMethod http://localhost:5000/api/response/playbooks` being executed, resulting in a list of five playbooks: BRUTE_FORCE, PORT_SCAN, SUSPICIOUS_PORT, EXTERNAL_TO_INTERNAL, and HIGH_TRAFFIC_VOLUME, each with a corresponding action object.

SOAR Response

Playbooks **5** Response Actions **0** Simulated Actions **0**

Response Activity

No response actions executed yet.

```
PS C:\Users\ADMIN\network-defense-sim> Invoke-RestMethod http://localhost:5000/api/response/playbooks

BRUTE_FORCE       : @{name=Brute Force Response; actions=System.Object[]}
PORT_SCAN         : @{name=Port Scan Response; actions=System.Object[]}
SUSPICIOUS_PORT   : @{name=Suspicious Port Response; actions=System.Object[]}
EXTERNAL_TO_INTERNAL : @{name=External Access Response; actions=System.Object[]}
HIGH_TRAFFIC_VOLUME : @{name=High Traffic Response; actions=System.Object[]}
```

Command / action for this screenshot

Invoke-RestMethod http://localhost:5000/api/response/playbooks

Invoke-RestMethod http://localhost:5000/api/response/actions

12. Execute Simulated Response

A selected alert can be passed to the response layer and a simulated response workflow can be executed.

SNAPSHOT 11 — Execute Simulated Response

```
id           : alert-1790169546243
timestamp    : 2026-09-23T13:19:06.243Z
sourceIP     : 10.0.0.30
destinationIP : 10.0.0.10
service      : HTTPS
protocol     : TCP
destinationPort : 443
action       : ALLOW
profile      : VPN Traffic
risk         : @{score=20; level=LOW}
detections   : {@{detected=True; rule=HIGH_TRAFFIC_VOLUME; severity=MEDIUM; score=20; message=Unusually
               high traffic volume detected; mitre=}, @{detected=False; mitre=}, @{detected=False;
               mitre=}, @{detected=False; mitre=}...}
correlations  : {}
eventCount   : 1
status       : OPEN
```

Command / action for this screenshot

```
$response = Invoke-RestMethod -Method POST -Uri http://localhost:5000/api/response/execute -
ContentType 'application/json' -Body (@{alertId=$alertId;incidentId=$incidentId} | ConvertTo-Json)
$response | Format-List
```

13. Incident Resolution

After investigation and response, the incident can be resolved with a documented resolution statement.

SNAPSHOT 12 — Incident Resolution

```
id          : INC-1790169930241
title       : Security Incident
description  : Security event requires investigation.
severity    : LOW
riskScore   : 20
status      : RESOLVED
priority    : P4
sourceIP    : 10.0.0.101
destinationIP : 10.0.0.10
detectionRules : {HIGH_TRAFFIC_VOLUME, $null, $null, $null...}
mitreTechniques : {@{techniqueId=T1049; technique=System Network Connections Discovery; tactic=Discovery;
description=Adversaries may inspect network connections to understand active network
communications.}}
relatedAlerts : {alert-1790169855090}
```

Command / action for this screenshot

Resolve the current incident through the UI. API alternative:

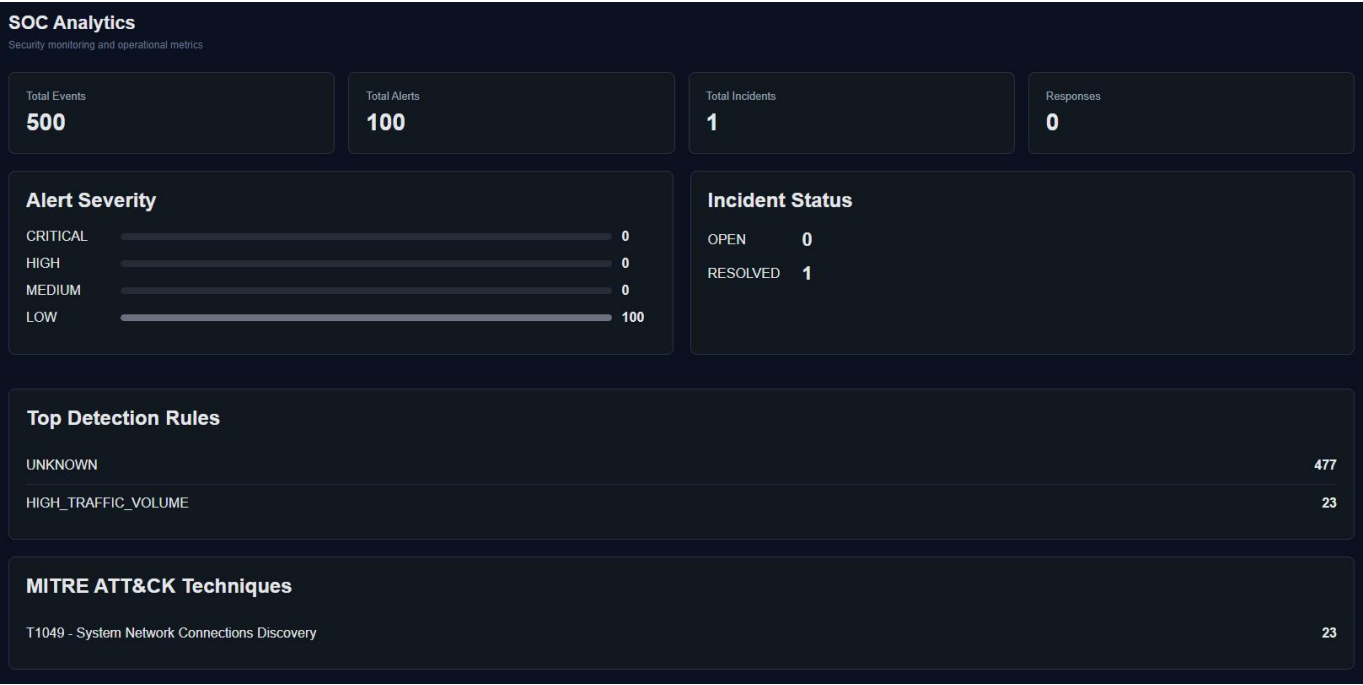
Invoke-RestMethod -Method POST -Uri "http://localhost:5000/api/incidents/\$incidentId/resolve" -

ContentType 'application/json' -Body (@{resolution='Investigation completed. Suspicious activity was analyzed and the simulated response workflow was executed.'} | ConvertTo-Json)

14. Analytics

Analytics aggregates security activity generated during the workflow for review of overall activity and trends.

SNAPSHOT 13 — Analytics



Command / action for this screenshot

Invoke-RestMethod <http://localhost:5000/api/analytics>

16. End-to-End SOC Workflow

The complete workflow demonstrated by the project is:

- 1. GENERATE — Controlled traffic/security activity is generated.
- 2. DETECT — Detection logic analyzes activity and produces alerts.
- 3. INVESTIGATE — The analyst reviews an alert and creates an incident.
- 4. RESPOND — A simulated SOAR playbook/response action is executed.
- 5. ANALYZE — Analytics summarizes the security activity.

SNAPSHOT 15 — Final Dashboard / Complete Workflow

